

Digital Systems Lab Final Assignment

Group-8

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1 Digital Clock with Menu

1.1 Introduction

The digital clock module implemented in Verilog features multiple modes and functionalities, including 24-hour/12-hour timekeeping, alarm setting, timer, stopwatch, and date management. Each of these modes is carefully managed with dedicated inputs for user control. This report will explain the clock's modes and the edge cases that are handled within the system to ensure robustness.

1.2 Modes of the Digital Clock

The digital clock supports several distinct modes, each designed to cater to specific functionalities. The clock operates based on the `menu_select` input, which allows the user to choose between different modes.

1.3 Time Mode

In the time mode (menu select 3'd0), the clock displays the current time. The time is updated every second (1Hz tick). The user can set the time via the `set_time` input, providing values for hours, minutes, and seconds. The clock supports both 12-hour and 24-hour formats, controlled by the `mode_24h` input. In the 12-hour format, the clock switches between AM and PM, while in the 24-hour format, it follows standard 24-hour timekeeping.

Edge Case: Midnight Transition

One of the key edge cases handled is the transition from 11:59:59 to 12:00:00. For the 12-hour mode, the system handles the switch between AM and PM. In the 24-hour mode, the hour resets to 00:00:00 after reaching 23:59:59.

Edge Case: Date Rollover

The clock also takes care of the date rollover at midnight. If the day reaches the end of the month (e.g., 31st in a month with 31 days), the date increments and the month advances. If the month reaches December (12th month), it resets to January (1st month) and the year increments by 1. The system also handles leap years for February, ensuring that it has 29 days when applicable.

1.4 Timer Mode

The timer mode (menu select 3'd1) allows the user to set a timer. The timer starts upon the `start_timer` input, and can be paused or resumed with the `pause_timer` input. The timer counts down in seconds. When the timer reaches zero, a buzzer is activated. The timer is controlled using a 1Hz tick to decrement the time.

Edge Case: Timer Expiry

When the timer expires, the timer stops, and the timer buzzer sounds for a brief period. The user can reset the timer or start a new one at any time.

1.5 Alarm Mode

In the alarm mode (menu select 3'd2), the user can set the alarm time using the `set_alarm` input. The alarm rings when the current time matches the set alarm time. The alarm buzzer remains active for 10 seconds once triggered.

Edge Case: Alarm Activation

The alarm is activated at the beginning of the minute (when both the hour and minute match the alarm settings and the second is zero). The system ensures that the alarm only rings once per day unless reset.

1.6 Date Set Mode

In the date set mode (menu select 3'd3), the user can set the day, month, and year of the clock. The date is updated via the `set_time` input, which is handled similarly to the time setting but with date-specific inputs. This mode allows for flexible date management, including handling different month lengths and leap years.

1.7 Stopwatch Mode

The stopwatch mode (menu select 3'd4) allows the user to start, stop, and reset the stopwatch. The stopwatch increments in hundredths of a second, seconds, and minutes. The user controls the stopwatch via the `sw_start`, `sw_stop`, and `sw_reset` inputs. The stopwatch updates with a 100Hz clock for precision.

Edge Case: Stopwatch Overflow

The stopwatch handles the overflow when the minutes reach 59 and resets to 0. Similarly, it resets the seconds after they reach 59, and the hundredths of a second after they reach 99. The system ensures that the stopwatch can run continuously without glitches.

1.8 Timing Diagram

```
timer=10 buzzer_state=0
timer= 9 buzzer_state=0
timer= 8 buzzer_state=0
timer= 7 buzzer_state=0
timer= 6 buzzer_state=0
timer= 5 buzzer_state=0
timer= 4 buzzer_state=0
timer= 3 buzzer_state=0
timer= 2 buzzer_state=0
timer= 1 buzzer_state=0
timer= 0 buzzer_state=0
timer= 0 buzzer_state=1
timer= 0 buzzer_state=1

Alarm set at 07:00:00
06:59:56 alarm_buzzer=0
06:59:57 alarm_buzzer=0
06:59:58 alarm_buzzer=0
06:59:59 alarm_buzzer=0
07:00:00 alarm_buzzer=0
07:00:01 alarm_buzzer=0
07:00:02 alarm_buzzer=1
07:00:03 alarm_buzzer=1
07:00:04 alarm_buzzer=1
07:00:05 alarm_buzzer=1
07:00:06 alarm_buzzer=1
07:00:07 alarm_buzzer=1
07:00:08 alarm_buzzer=1
07:00:09 alarm_buzzer=1
07:00:10 alarm_buzzer=1
07:00:11 alarm_buzzer=1
07:00:12 alarm_buzzer=0
07:00:13 alarm_buzzer=0
07:00:14 alarm_buzzer=0
07:00:15 alarm_buzzer=0

Setting date to 28-2-2024
Before setting -
Date = 11:1-2020
After setting -
Date = 30:6-2024

Setting time after setting date
23:59:51 30-06-2024
23:59:52 30-06-2024
23:59:53 30-06-2024
23:59:54 30-06-2024
23:59:55 30-06-2024
23:59:56 30-06-2024
23:59:57 30-06-2024
23:59:58 30-06-2024
23:59:59 30-06-2024
00:00:00 01-07-2024
00:00:01 01-07-2024
```

```
Resetting date to 1-1-20 after 30-4-25
23:59:56 30-04-2025
23:59:57 30-04-2025
23:59:58 30-04-2025
23:59:59 30-04-2025
00:00:00 01-01-2020
00:00:01 01-01-2020
00:00:02 01-01-2020
00:00:03 01-01-2020
00:00:04 01-01-2020
00:00:05 01-01-2020

28-2 to 1-3 during non-leap years
23:59:56 28-02-2023
23:59:57 28-02-2023
23:59:58 28-02-2023
23:59:59 28-02-2023
00:00:00 01-03-2023
00:00:01 01-03-2023
00:00:02 01-03-2023
00:00:03 01-03-2023
00:00:04 01-03-2023
00:00:05 01-03-2023

28-2 to 29-2 during leap years
23:59:56 28-02-2020
23:59:57 28-02-2020
23:59:58 28-02-2020
23:59:59 28-02-2020
00:00:00 29-02-2020
00:00:01 29-02-2020
00:00:02 29-02-2020
00:00:03 29-02-2020
00:00:04 29-02-2020
00:00:05 29-02-2020

29-2 to 1-3 during leap years
23:59:56 29-02-2020
23:59:57 29-02-2020
23:59:58 29-02-2020
23:59:59 29-02-2020
00:00:00 01-03-2020
00:00:01 01-03-2020
00:00:02 01-03-2020
00:00:03 01-03-2020
00:00:04 01-03-2020
00:00:05 01-03-2020

Stopwatch for 2s
00:02:00
```

```
VCD info: dumpfile gtk.vcd opened for output.

24-hour clock
23:59:51 01-01-2020
23:59:52 01-01-2020
23:59:53 01-01-2020
23:59:54 01-01-2020
23:59:55 01-01-2020
23:59:56 01-01-2020
23:59:57 01-01-2020
23:59:58 01-01-2020
23:59:59 01-01-2020
00:00:00 02-01-2020
00:00:01 02-01-2020
00:00:02 02-01-2020
00:00:03 02-01-2020
00:00:04 02-01-2020

12-hour clock
11:59:50 AM
11:59:57 AM
11:59:58 AM
11:59:59 AM
12:00:00 PM
12:00:01 PM
12:00:02 PM
12:00:03 PM
12:00:04 PM
12:00:05 PM
```

